BIOMEDICAL LIMITED, ILORIN, No 1 Ohimege Road, Industrial Estate, Ilorin Kwara State, PMB 1499.	RAW MATERIAL ANALYTICAL METHODS	STP №	: BML/QA/STP/RMA/005/F02
		Material Code №	: RMA005
		Effective Date	: 4th Jan., 2024
		Supersedes	: 00
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	SODIUM LACTATE	Revision No	: 01
		Ref. Book	BP 2023/IN-HOUSE
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Storage Conditions:	**Store in a well-closed container, protected from light**	**Department**	**: Quality**
Retest Frequency:	**12 months from date of release**	**Section**	**Quality Control**

ANALYTICAL METHODS:

1 **DESCRIPTION**

1.1 Physically examine a representative sample of the material for Appearance and other characteristics: A clear, colourless, slightly syrupy liquid, miscible with water and with alcohol.

2 **IDENTIFICATION**

2.1 Sodium Test

2.1.1 Preparation Of Potassium Antimonate V Solution

2.1.2 Weigh about 2g of potassium antimonite V into a 250ml beaker.

2.1.3 Add 95ml of hot water, drop a bar magnet and mix (using a magnetic stirrer) till dissolution is complete

2.1.4 Cool rapidly to room temperature in a water-bath at 25°C

2.1.5 Add 50ml of 5%w/v solution of potassium hydroxide and 1ml of 2M sodium hydroxide and mix.

2.1.6 Allow standing for 24 hours, filter through Whatman #1 filter-paper into another 250ml beaker

2.1.7 Dilute to 150ml with distilled water and mix.

2.2 Procedure

2.2.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

2.2.2 Add 60ml of distilled water and mix until the sample dissolve

2.2.3 Dilute to the mark with distilled water and mix.

2.2.4 Pipette 2ml of the solution into a test tube.

2.2.5 Add 2ml of 15% w/v solution of potassium carbonate and heat to boiling; No precipitate is produced.

2.2.6 Add 4ml of potassium antimonite V and heat to boiling.

2.2.7 Allow cooling in ice and if necessary scratch the inside of the tube with a glass rod; a dense, white precipitate is produced

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		

2.3 Lactate Test

2.3.1 Bromine Water Preparation

2.3.2 Using measuring cylinder, measure 100ml of distilled water into Amber 500ml Reagent Bottle.

2.3.3 Carefully pipette 3.0ml of Bromine into the above reagent bottle, cover and vigorously shake for 2 minutes. Then allow separation before pipetting the bromine water.

2.4 NOTE:

2.4.1 Before use, shake occasionally during 24 hours.

2.4.2 Do not pipette Bromine with mouth!!!

2.4.3 Handle all bromine solution in the fumehood.

Store the solution over an excess of bromine, protected from light.

2.5 Acidified Sodium Nitroprusside

2.5.1 Accurately weigh and transfer about 10g of sodium nitroprusside into a 100ml volumetric flask.

2.5.2 Add 60ml of 1 M sulphuric acid solution and mechanically shake/sonicate till dissolution is complete.

2.5.3 Dilute to the mark with same and mix.

2.6 Procedure

2.6.1 Accurately weigh and transfer about 5g of sample into a 100ml volumetric flask.

2.6.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

2.6.3 Dilute to the mark with distilled water and mix.

2.6.4 Pipette 5ml of this solution into a 50ml beaker.

2.6.5 Add 1ml of bromine water, 0.5ml of 1M sulphuric acid and heat on the water-bath until the colour is discharged, stirring occasionally with a glass rod.

2.6.6 Add 4 g of ammonium sulphate, mix till dissolution is complete.

2.6.7 Dropwisely add, without mixing, 0.2 ml of a 10% w/v solution of sodium nitroprusside [sodium hexacyanoferrate (II) or sodium ferrocyanate] in 1M

sulphuric acid.

2.6.8 Add dropwise, without mixing, 0.2 ml of acidified sodium nitroprusside solution.

2.6.9 Still without mixing add 1 ml of concentrated ammonia and allow standing for 30 minutes: A dark green ring is produced at the interface of the two liquids.

3 APPEARANCE OF SOLUTION/PH

3.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

3.2 Add 60ml of distilled water, mechanically shake/sonicate till dissolution is complete.

3.3 Dilute to the mark with distilled water and mix: Solution is clear and not more intensely coloured than reference solution BY7 – see appendix 1.

3.4 After 4 hours of preparation of sample solution, calibrate a pH meter with buffer pH 4 followed by buffer pH 7, then determine the pH of 25ml of sample solution: **pH of the solution is between 6.5 – 9.0**

4 Sucrose and reducing sugars

4.1 Sodium Sulphide solution

4.1.1 Weigh about 12g of sodium sulphide into a 250ml beaker

4.1.2 Add 45ml of 75%v/v glycerol solution.

4.1.3 Transfer to a hotplate and heat with stirring until complete dissolution of the sulphide.

4.1.4 Allow cooling to room temperature, transfer to 100ml volumetric flask, dilute to the mark with 75%v/v glycerol solution and mix.

4.2 Fehling's Solution I

4.2.1 Dissolve, dilute 34.6g of copper (II) sulphate to 500.0ml with distilled water and mix.

4.3 Fehling's Solution II

4.3.1 Into a 1000ml beaker, add 173g of potassium sodium tartrate, 50g of sodium hydroxide and 400ml of distilled water

4.3.2 Using a suitable stirrer, mix till dissolution is complete.

4.3.3 Dilute to the mark with distilled water and mix.

4.4 Procedure

4.4.1 Weigh about 0.5g of sample into a 250ml beaker

4.4.2 Add 10ml of distilled water and heat on water-bath operating at 60°C for 30 minutes, swirling from time to time to effect dissolution.

4.4.3 Add 1ml 0f 50%v/v ammonia and mix.

4.4.4 Add 1ml of sodium sulphide solution, mix and leave for 30 minutes

4.4.5 Filter through Whatman #1 filter paper into a 250ml conical flask.

4.4.6 Wash the precipitate with two quantities, each of 5 ml, of distilled *water*

4.4.7 Add 10ml of 2M hydrochloric acid

4.4.8 Boil until the volume is reduced to about 10 ml.

4.4.9 Cool, add 15 ml of 10.6%w/v sodium carbonate solution, allow standing for 5 minutes and filter through Whatman #1 filter paper into a 250ml conical flask

4.4.10 Transfer the filtrate into a 100ml volumetric flask, dilute to the mark with distilled water and mix

4.4.11 Pipette 5ml of this solution into a test tube.

4.4.12 Add 1.0ml each of Fehling's Solutions I and II.

4.4.13 Boil solution for 1 minute and allow standing for 1 minute: **No red precipitate is formed.**

5 SULPHATES.

5.1 SULPHATE STANDARD PREPARATION (150ppm)

5.1.1 Accurately weigh and transfer about 0.905g of analytic grade potassium sulphate into a100ml volumetric flask.

5.1.2 Add 60ml of 30% v/v ethanol in water and sonicate till dissolution is complete.

5.1.3 Dilute to the mark with 30% v/v ethanol in water and mix "(1000ppm sulphate standard stock solution).

5.1.4 Pipette 1.0ml of this solution into another 100ml volumetric flask, dilute to the mark with 30% v/v ethanol in water and mix (150ppm sulphate working standard).

5.2 Sample Preparation

5.2.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

5.2.2 Add 60ml of distilled water, mechanically shake/sonicate till dissolution is complete.

5.2.3 Dilute to the mark with distilled water and mix.

5.3 Procedure

5.3.1 Pipette 7.5ml of sample solution and 3.0ml of the standard solution into separate 25ml volumetric flasks.

5.3.2 Add to each flask 1ml of acetic acid and mix.

5.3.3 To another flask, add 13.0ml of distilled water, 2ml of 150ppm sulphate standard solution, and 1ml of acetic acid into 25ml volumetric flasks

5.3.3 To each flask, add 1ml of 25% w/v solution of barium chloride solution and 1ml of 2M nitric acid.

5.3.4 Dilute to the mark with 30% v/v ethanol in water and mix.

5.3.5 Shake each flask and allow standing for 5 minutes: **Any opalescence produced is not more intense than that of Sulphate standard (10ppm SO_4).**

6 CHLORIDES

6.1 Reagents

6.1.1 250ppm Chloride Standard Solution

Accurately weigh and transfer about 412.3mg of analytical grade sodium chloride into a one-litre volumetric flask.

Add 600ml of distilled water, mechanically shake/sonicate till dissolution is complete. than that in the Standard (250ppm or 0.25%)

Dilute to the mark with distilled water and mix.

Pipette 5.0ml of this solution into a 100ml volumetric flask, dilute to the mark with distilledwater and mix (5ppm chloride standard solution).

6.1.2 2M nitric acid

Pipette 12.6ml of conc Nitric acid to 100ml volumetric flask, dilute to the mark with distilled water and mix

6.1.3 4.25% w/v silver nitrate

Accurately weigh and transfer about 4.25g to 100ml volumetric flask, dilute to the mark with distilled water and mix

6.2 Sample Preparation

6.2.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

6.2.2 Add 60ml of distilled water, mechanically shake/sonicate till dissolution is complete.

6.2.3 Dilute to the mark with distilled water and mix.

6.3 Procedure

6.3.1 Pipette 5.0ml each of sample and 15.0ml of the Chloride standard preparations into separate 50ml volumetric flasks.

6.3.2 To each, add 1.0ml of 2M nitric acid and 1.0ml of 4.25%w/v (0.1N) silver nitrate solution, dilute to the mark with distilled water and mix.

6.3.3 After 15 minutes, compare the opalescence of both sample and standard against a black background***: After 15 minutes, any opalescence in the sample solution is not more intense than in the reference standard preparation***

7 HEAVY METALS

7.1 Lead stock solution(100ppm)

7.1.1 Accurately weigh and transfer about 160mg of lead into 250ml beaker.

7.1.2 Add 100ml of de-ionized water, 1ml of nitric acid and mix till dissolution is complete.

7.1.3 Transfer into a 1-litre volumetric flask.

7.1.4 Rinse the 250ml beaker with several quantities of de-ionized water into the 1-litre volumetric flask.

7.1.5 Dilute with distilled water to 1000ml and mix. Prepare and store this solution in glass containers free from soluble lead salts.

7.3 Lead stock solution (1ppm)

7.3.1 Pipette 2.0ml of the 100ppm lead standard solution into a 100ml volumetric, dilute to the mark with distilled water and mix.

7.4 Buffer Solution pH3.5

7.4.1 In a 250ml beaker, dissolve 25.0g of ammonium acetate in 25ml of water and add 38.0ml of 7M hydrochloric acid.

7.4.2 Adjust the pH if necessary with 2M hydrochloric acid or 6M ammonium hydroxide and dilute to 100.0ml with water

7.5 Thioacetamide SOLUTION(4%)

7.5.1 Dissolve 4g of thioacetamide in a 100ml volumetric flask, make up with distilled water

7.5.2 Thioacetamide Reagent
To 2ml of thioacetamide solution in a 100ml volumetric flask, add 7.5ml of water, 37.5ml of M sodium hydroxide and 50ml of glycerol (85%). Heat in a water-bath for 20 seconds. Prepare immediately before use.

7.6 Hot Concentrated Hydrochloric Acid

7.6.1 Transfer about 50ml of concentrated hydrochloric acid into 100ml beaker.

7.6.2 Place on a steam-bath and heat for about 20 minutes.

7.7 Hydrochloric Acid Saturated Methyl Isobutyl Ketone

7.7.1 Transfer 100ml of Methyl Isobutyl Ketone into a 250ml separating funnel.

7.7.2 Add 1ml of concentrated hydrochloric acid, cover and vigorously shake for 2 minutes.

7.8 Sample Preparation

7.8.1 Weigh about 5g of sample and 1g of magnesium oxide into a mortar.

7.8.2 Using a suitable spatula, thoroughly mix.

7.8.3 Accurately weigh and transfer about 3g of thoroughly mixed sample in step ‘ii’ into a porcelain crucible.

7.8.9 Transfer into furnace operating at 800°C and ignite to dull redness until a homogeneousmass is obtained. Heat further for about 1 hour.

7.8.10 Dissolve the residue in the crucible with 4 X 5ml of hot concentrated hydrochloric acid transferring all the extract into a 250ml separating funnel. Then allow cooling to room temperature.

7.8.11 Extract the content of the separator with 3 X 20ml hydrochloric acid saturated methyl isobutyl ketone, shaking for three minutes each time.

7.8.12 Allow standing and filter the lower aqueous layer through Whatman #1 filter paper into50ml beaker.

7.8.13 Transfer the 50ml beaker onto a steam-bath and heat until the volume is reduced to about10ml.

7.8.14 Allow cooling to room temperature, dilute to the 25ml-mark with distilled water and mix.

7.8.15 Under the monitoring of a pH meter, adjust the pH to pH 7 using 50%v/v ammonia solution

7.8.16 Transfer the solution into a 50ml volumetric flask, dilute to 50ml-mark with distilled water and mix.

7.8.17 To 12ml of the solution thus obtained (in a test tube), add 2ml of buffer solution pH3.5,1.2ml of thioacetamide reagent and immediately mix.

7.9 <u>Standard Preparation</u>

7.9.1 Transfer 10 ml of 1ppm lead standard solution and 2 ml of the sample solution into a suitable test tube.

7.9.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

7.10 Blank Preparation

7.10.1 Prepare a blank using a mixture of 10ml of water and 2ml of sample solution.

7.10.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

7.11 OBSERVATION

7.11.1 Compared to the blank, after 2 minutes any brown colour in the solution obtained with the sample is not more intense than in the standard solution

8 OXALATES

8.1 Calcium Chloride Solution

8.1.1 Accurately weigh and transfer about 7.35g into 100ml volumetric flask.

8.1.2 Add 60ml of distilled water, mechanically shake/sonicate till dissolution is complete.

8.1.3 Dilute to the mark with distilled water and mix.

8.2 Procedure

8.2.1 Accurately weigh and transfer about 0.6g of sample 50ml beaker.

8.2.2 Add 40ml of distilled water, 10ml of 2M sulphuric acid swirl till dissolution is complete.

8.2.3 Transfer the dissolved sample into a 150ml separating funnel, add 50ml of diethyl ether and shake 5 minutes

8.2.4 Transfer the lower aqueous layer into another 150ml separating funnel, add 20ml of diethyl ether and shake for 5 minutes

8.2.5 Collect all the ethereal layers together in a 250ml beaker and evaporate to dryness.

8.2.6 Add 15ml of distilled water to the residue in the beaker and swirl for 3 minutes to dissolve the residue.

8.2.7 Filter the content of the beaker through Whatman #1 filter paper into a 100ml beaker.

8.2.8 Transfer the 100ml beaker and filtrate to a hotplate and heat until the volume is reduced to5 ml.

8.2.9 Add 1ml of 2M acetic acid, 1.5 ml of calcium chloride solution and mix.

8.2.10 Allow standing for 30 minutes: **No precipitate is formed.**

9 IRON

9.1 20ppm Iron Standard Stock

9.1.1 Accurately weigh and transfer about 140.4mg of Ferrous Ammonium Sulphate hexahydrate[Fe(NH4) 2 (SO4) 2.6H2O] into a one-litre volumetric flask.

9.1.2 Add 300ml of distilled water, 25ml of concentrated sulphuric acid, mechanically shake/sonicate till dissolution is complete.

9.1.3 Dilute to the mark with distilled water and mix. KEEP THIS SOLUTION FOR NOT MORE THAN 3 MONTHS!

9.2 Sample Preparation

9.2.1 Accurately weigh and transfer about 20g of sample into a 100ml volumetric flask.

9.2.2 Add 60ml of distilled water, mechanically shake/sonicate till dissolution is complete.

9.2.3 Dilute to the mark with distilled water and mix.

9.3 Procedure

9.3.1 Pipette 5.0ml of the sample and 15.0ml of standard preparations into separate 50ml volumetric flask.

9.3.2 Add 2ml of 20%w/v citric acid, 0.1ml of mercpatoacetic acid and mix.

9.3.3 Dilute to the mark with distilled water, mix and allow standing for 10 minutes.

9.3.4 Using a suitable measuring cylinder, transfer 40ml of each preparation into separate Nessler cylinders and compare the intensity of the colour developed: **Pink colouration obtained in the test solution is not more intense than the standard solution.**

10 LOSS ON DRYING

10.1 Weigh two labelled petri dishes (W1) that has previously been dried at 105°C and cooled

10.2 Accurately weigh about 0.5g onto a petri-dish, ensuring they are evenly spread, then determine their weight together (W2).

10.3 Transfer into an Oven operating at 105°C

10.4 Dry for 5 hours.

10.5 Transfer into a desiccators to cool to room temperature.

10.6 Weigh the petri-dish with its content (W3).

Calculation:

Calculate the moisture content as follows:

$$\% \text{ Moisture} = \frac{(W2 - W3) \times 100}{W2 - W1}$$

Where,

W1 = weight of empty petri-dish, g

W2 = weight of sample and petri-dish, g.

W3 = weight of petri-dish and the dried sample, g

11 BARIUM

11.1 Sample Preparation

11.1.1 Weigh about 20g of sample into a 100ml volumetric flask

11.1.2 Add 60ml of distilled water, mechanically shake/sonicate till dissolution is complete.

11.1.3 Dilute to the mark with distilled water and mix.

11.1.4 Procedure

11.1.5 Pipette 10.0ml of this solution and 1ml of 1M Sulphuric acid solution into a test tube and mix:

11.1.6 Pipette 10.0ml of this solution and 1ml of distilled water into another test tube and mix:

11.1.7 Allow 15 minutes and compare the turbidity of the two solutions: After at least 15 minutes, any opalescence in the sample solution plus distilled water is not more intense than that in a mixture of 10 ml of sample solution and 1ml of 1M Sulphuric acid solution

12 ASSAY METHOD

12.1 **Reagents**

12.1.1 0.1M perchloric acid

12.1.2 0.1% crystal violet in glacial acetic acid
Accurately weigh and transfer about 0.1g to 100ml volumetric flask, dilute to the mark with glacial acetic acid and mix

12.2 Procedure

12.2.1 Accurately weigh and transfer about 75mg of sample into each into two separate 250ml conical flasks.

12.2.2 Add 10ml of glacial acetic acid and 20ml of acetic anhydride swirl until dissolution is complete.

12.2.3 Allow standing for 15 minutes, add 0.1ml of 0.1%w/v crystal violet and titrate with 0.1M perchloric acid to an emerald green endpoint

$$\% \text{ Sodium Lactate)} = \frac{\text{Sample Titre x Standard factor of } HClO_4 \text{ X 11.21mg x 100}}{\text{Sample Weight (mg)}}$$

13 ALUMINIUM

13.1 Acetate Buffer pH 6.0

13.1.1 Weigh about 40g of ammonium acetate into a 250ml beaker

13.1.2 Add 100ml of distilled water, 1.6ml of glacial acetic acid and mix till dissolution is complete.

13.1.3 Adjust the pH to 6.0, if necessary, using 10M ammonia or 5M acetic acid

13.1.4 Dilute to the 200ml- mark with distilled water and mix.

13.2 200ppm Aluminium Standard Solution

13.2.1 Weigh about 0.352g of [Aluminium Potassium Sulphate (AlK(SO4)2,12H2O) into a 100ml volumetric flask.

13.2.2 Add 30ml of distilled water, 10 ml of 1M sulphuric acid and mechanically shake/sonicate till dissolution is complete.

13.2.3 Dilute to the 100ml- mark with distilled water and mix. **Store this solution for not**

more than 3 months.

13.3 2ppm Aluminium Standard Solution **(Prepare Immediately Before Use!)**

13.3.1 Pipette 1.0ml of 200ppm aluminium solution into 100ml volumetric flask.

13.3.2 Dilute to the 100ml- mark with distilled water and mix. KEEP THIS SOLUTION FOR NOT MORE THAN 3 MONTHS!,

13.4 0.5% w/v Solution Of 8-Hydroxyquinoline

13.4.1 Weigh and transfer about 1g of 8-hydroxyquinoline into a dry 200ml volumetric flask.

13.4.2 Add 100ml of chloroform, mechanically shake/sonicate till dissolution is complete.

13.4.3 Dilute to the 200ml- mark with chloroform and mix.

13.4 Sample Preparation

13.4.1 Accurately weigh and transfer about 4g of sample into a 100ml volumetric flask.

13.4.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

13.4.3 Add 10ml of acetate buffer pH 6.0, dilute to the mark with distilled water and mix.

13.5 Procedure

13.5.1 Into three 250 ml Separate funnels, pipette and add in the following order:

Reagent s	Sample	Standard	Blank
Sample Solution	100.0ml	-	-
2ppm Aluminium Standard Solution	-	2.0ml	-
Water	-	98.0ml	100.0ml
Acetate Buffer pH 6.0	10.0ml	10.0ml	10.0ml

13.5.2 Separately extract each of the sample, standard and blank preparations in the above table with successive quantities of 20, 20 and 10 ml of a 0.5% w/v solution of 8-hydroxyquinoline in chloroform, filtering the extracts through chloroform-wet

Whatman #1 filter paper into separate, 'sample', 'standard', 'blank' labeled, dry 50ml volumetric flasks

14 Microbiological Tests

14.1 <u>Limulus Amebocyte Lysate LAL Test</u>

14.1.1 Open the seals of the test tubes of lyophilized LAL Reagents

14.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

14.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

14.1.4 Incubate the reaction mixture immediately at 37°c for 60minutes. Avoiding vibration during incubation.

14.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

14.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

14.1.7 Read reaction tubes one by one.

14.1.8 Invert the tube in one smooth motion.

14.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by

the absence of solid clot after inversion.

Testing Methodology of Bacterial Endotoxin (LAL)

14.2.1 Each specimen shall be tested in duplicate

14.2.2 NMC: Take 100µl(0.1ml) LRW

14.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

14.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

14.2.5 Add the LAL reagents as mentioned below in the table 1

14.2.6 Mix gently and incubate the tubes in heating block at 37^0C ± 1 for 60min.

14.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 180^0C and note the observations.

14.2.8 If a firm gel has formed, record the results as positive.

14.2.9 A result is negative, if an intact gel is not formed.

14.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

Pyrogen

14.3 14.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed.

14.3.2 Withhold all food from the rabbits used during the period of the test

14.3.3 Draw out the required number of rabbits from their living room according

to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring.

14.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit.

14.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm.

14.3.6 If the initial temperature is less than 38.8^{0}c or more than 39.8^{o}cexclude the animal from the test.

14.3.7 Ensure all the needles and syringe to be used are new

14.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 1^{0}c.

14.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection.

14.3.10 Any rabbits showing more than 0.2^{0}c temperature difference between two successive readings in the last three readings can be excluded from the test.

14.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes.

14.3.12 Inject the solution intended to be tested to the marginal veins of the

rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution.

14.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected.

14.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

14.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution

14.4 **MLT**

14.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

14.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

14.4.3 Testing to be performed under LAF of specified area

14.4.4 Arrange all the required material on the LAF before starting the test.

14.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

14.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45°C) in two petri plates for TAMC

14.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45°C) in two petri- plates for TYMC

14.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate

14.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45°c and allow for solidification.

14.4.1 Incubate the plates at 30°C to 35°C for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 20°C to 25°C for 5 days in an inverted position.